

UV Radiation Sensor (AS7331)

Hardware Overview

What is it?

The **AS7331** is a high-precision digital sensor designed to measure Ultraviolet (UV) radiation. It splits the light it sees into the three main UV bands: **UV-A** (blacklight/aging), **UV-B** (sunburn), and **UV-C** (germicidal/sterilization), giving you scientific-grade measurements of light intensity.

Electrical Explanation

Inside the chip are microscopic optical filters made of specially colored glass. These filters block out visible light and only let specific UV wavelengths pass through to a tiny photodiode.

- **Irradiance:** It measures the absolute intensity of the UV light hitting it in microWatts per square centimeter ($\mu\text{W}/\text{cm}^2$).
- **Digital Precision:** It processes the tiny electrical charges built up on the photodiodes internally using a high-resolution ADC, converting them into a stable digital signal.

Key Hardware Facts:

- **Connection:** I²C Bus (SDA/SCL).
- **Default Address:** 0x74.
- **Voltage:** Operates at 3.3V to 5V.
- **Signal Flow:** UV Light → Optical Filter → Photodiode → ADC → I²C Data → Processor.

Software & Programming

kcomp_AS7331.h

This library handles the internal state machine of the sensor and automatically applies the mathematical compensation formulas to turn raw values into $\mu\text{W}/\text{cm}^2$ floating-point numbers.

Usage: `#include <kcomp_AS7331.h>`

Function Reference

Function Name	Description & Parameters
<code>bool initAS7331()</code>	Setup: Starts I ² C and sets up a default continuous measurement loop. ▪ Returns <code>true</code> if sensor is connected, <code>false</code> if missing.
<code>void setAS7331Gain(uint8_t gainReg)</code>	Set Sensitivity: Changes the sensor's hardware amplification. ▪ <code>gainReg</code>: Use one of the built-in macros: <code>AS7331_GAIN_2048X</code> (Highest sensitivity - Dark black room) <code>AS7331_GAIN_256X</code> (High sensitivity - Indoor lighting) <code>AS7331_GAIN_64X</code> (Default - Mixed lighting) <code>AS7331_GAIN_16X</code> (Low sensitivity - Direct sunlight) <code>AS7331_GAIN_1X</code> (Lowest sensitivity)
<code>float getAS7331_UVA()</code>	Read UV-A: UV-A intensity in $\mu\text{W}/\text{cm}^2$. (Passes through glass).
<code>float getAS7331_UVB()</code>	Read UV-B: UV-B intensity in $\mu\text{W}/\text{cm}^2$. (Mostly blocked by glass).
<code>float getAS7331_UVC()</code>	Read UV-C: UV-C intensity in $\mu\text{W}/\text{cm}^2$. (Blocked by the ozone layer).

⚠ Pitfalls & Troubleshooting

- **Why are my readings so weird? (Light Physics)**

UV light interacts with the environment very differently than visible light!

Near Zero Readings: You might read $0.05 \mu\text{W}/\text{cm}^2$ even in a bright room. Window glass acts like a brick wall for UV radiation. It blocks almost 100% of UV-B and UV-C.

Outside in the Sun (UV-C > 0): The Earth's ozone layer blocks natural UV-C from the sun, so UV-C should be 0.00. However, if you point the sensor at the sun, you might see $\sim 10.00 \mu\text{W}/\text{cm}^2$. This is called "Spectral Leakage." The sun is so bright that a fraction of visible light "bleeds" through the physical glass filter on the chip and tricks the sensor.

Sample Program

This program initializes the sensor, adjusts the sensitivity, and prints UV radiation data to the Serial Monitor.

```
#include <kcomp_AS7331.h>

void setup() {
  Serial.begin(9600);
  delay(1000);

  // 1. Initialize the sensor hardware
  if (!initAS7331()) {
    Serial.println("Error: AS7331 not found. Check I2C wiring!");
    while(1); // Stop here
  }

  Serial.println("AS7331 Online!");

  // Optional: Set Gain (Sensitivity)
  setAS7331Gain(AS7331_GAIN_64X);
}

void loop() {
  // 2. Read Values
  float uva = getAS7331_UVA();
  float uvb = getAS7331_UVB();
  float uvc = getAS7331_UVC();

  // 3. Output results
  Serial.print("UVA: "); Serial.print(uva); Serial.print(" uW/cm2 | ");
  Serial.print("UVB: "); Serial.print(uvb); Serial.print(" uW/cm2 | ");
  Serial.print("UVC: "); Serial.print(uvc); Serial.println(" uW/cm2");

  // Wait 1 second before taking the next reading
  delay(1000);
}
```

💡 **Try it out:** You can find the full sample program **SimpleUVSensorAS7331** in the K-Comp examples folder!